

**Textbook & Materials:**

- ☐ Required Text: Engineering Mechanics: Dynamics, Hibbeler, Fifteenth Edition, Pearson (2021).
- ☐ Access to MATLAB. MATLAB is available on department computers or can be accessed through ThunderApps. However, you are encouraged to obtain a student license for your personal computer.
- ☐ Lab tools and equipment access. All students registered for the class will have keycard access to the Wheaton Engineering Makerspace (WEM) in the basement of Meyer. Students must be trained on safety and operation before usage of the lab space, tools, and equipment.

**Course Description and Prerequisites:**

Topics include: kinematics and kinetics of particles; Newton's laws of motion; energy, momentum, systems of particles; rigid bodies; free-body diagrams; mass, acceleration, and force; plane motion of rigid bodies; and, conservation of energy and momentum. Prerequisite: ENGR 211. Pre or Co-requisite: PHYS 322 or MATH 333.

**Course Objectives:**

After satisfactory participation in this course, students should expect to be able to identify with the following course-level learning outcomes, as well as concept mastery objectives and skill development objectives.

*Course-Level Learning Outcomes:*

- A. I will be able to define, model, and analyze dynamic systems using engineering mechanics, mathematical and physical principles.
- B. I will be able to design, model, build, and analyze dynamic mechanical systems for a given problem that meets defined requirements.
- C. I will be able to apply multiple techniques to solve engineering mechanics problems (analytical, computational, empirical).
- D. I will be able to autonomously learn mechanics concepts, apply them, and self-assess and improve my understanding.
- E. I will be able to apply mechanics principles to solve engineering problems in various engineering disciplines.

*Concept Mastery Objectives:*

- 1. I will be able to calculate forces, positions, velocities, and accelerations in different coordinate systems using different mechanics methods.
- 2. I will be able to generate a set of relevant and complete free body diagrams for a given system in 2D and 3D.
- 3. I will be able to describe the motion of rigid bodies and points relative to a moving reference frame.
- 4. I will be able to calculate the energy and work done by a force or torque.
- 5. I will be able to describe the combined relative motion of interacting components in a mechanical system.
- 6. I will be able to describe the motion of vibrating systems subjected to forced vibrations and damping.

*Skill Development Objectives:*

- 1. I will be able to use modern engineering tools to solve mechanics problems.
- 2. I will be able to define the parameters, idealizations, approximations, and assumptions in undefined dynamics problems and explain their implications.
- 3. I will be able to apply dynamics principles to solve an open-ended problem.

4. I will be able to model a dynamic system using computational tools.
5. I will be able to optimize a design to meet given requirements.
6. I will be able to effectively collaborate within a team to identify, research, and solve a problem.

### Grading:

This course emphasizes collaboration, student engagement, and personal ownership of the learning process. Classwork, homework, projects, and assessments are therefore meant to encourage effort and collaboration rather than just correct answers.

### Homework

- ☐ Homework must be neat:
  - o 8.5 x 11" computer paper or 5x5 graph paper is highly encouraged.
  - o Please do not use non-perforated spiral bound paper.
  - o Header:

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[Assignment # & Due Date]

[Your Name] Pg ##

- o Diagrams must be carefully and neatly drawn (use of straight edge is highly encouraged).
- o Answers must be boxed or underlined.
- o All pages must be *stapled together* (Not dog-eared, paper-clipped, origami, bolted, etc.)
- o Neatness is important both in helping to facilitate problem solving, and in professionalism. Illegible work will receive reduced credit.
- o Computation done with MATLAB or other programs is allowed, but work must be turned in as attachments (commented code, figures, etc.).

- ☐ *Late Assignment Policy:*

Assignments, project deliverables, assigned readings and reflection questions must be completed and submitted by the due date and time. Late assignments will not be accepted without first meeting with the

### Unit Exams

- ☐ There are 3-unit exam assessments.
- ☐ Unit exam topics and tentative dates are indicated on the course schedule.

### Project

- ☐ The course project will be team-based and requires utilizing concepts learned from the course to solve an engineering problem. This project will emphasize practice with dynamics analysis, modeling, mechanisms, CAD, prototyping, and report writing.
- ☐ Details and requirements for the final project will be communicated during class.

### Final Exam

- ☐ The cumulative final exam will take place at the end of the semester and will cover the entire course material.

### Class Participation & Professionalism:

- ☐ Participation in a professional manner requires being on time, being attentive when others are speaking, contributing questions and comments, and having genuine respect for others. Attendance is considered mandatory in this course. Missing a class, arriving late, or leaving excessively early without a pre-approved excuse will result in a deduction in your final grade.
- ☐ Additionally, refrain from non-class-related uses of technology during class time. Give the instructor, your classmates, and yourself, your complete presence during class hours. Professionalism is also required in

communication with the course instructor and your peers (always use professional and respectful language in emails, Canvas inbox, etc.)

### Grading Scale:

|                                   |      | Letter Grade | Lower Cutoff | Letter Grade | Lower Cutoff |
|-----------------------------------|------|--------------|--------------|--------------|--------------|
| Homework                          | 20%  | A            | 93%          | C+           | 77%          |
| Participation/<br>Professionalism | 5%   | A-           | 90%          | C            | 73%          |
| Unit Exams                        | 30%  | B+           | 87%          | C-           | 70%          |
| Project                           | 30%  | B            | 83%          | D            | 60%          |
| Final Exam                        | 15%  | B-           | 80%          | F            | 0%           |
| Total                             | 100% |              |              |              |              |

### Engineering Lab and Workshop:

Engineering Lab safety training is available on CanvasX. All students must train on general laboratory and tool safety guidelines prior to usage of any equipment. Our Engineering Lab Supervisor [REDACTED] and Engineering Support TAs are available to help support project activities and provide equipment training and supervision. Their [REDACTED]

tools, basic Solidworks, 3D printing, CNC machines and general project guidance.

### Attendance & Classroom Policies:

As this is a collaborative, project-based course, it is important that you are present and fully engaged during class periods. *Attendance is considered mandatory in this course.* Leaving before the class session is over, or being excessively late will also count as an absence. Always bring your textbooks and project related materials.

### Class Communication:

Course assignments, announcements and documents will be communicated in class, via email, or through Canvas. Grades will be posted to Canvas or returned in class. Additionally, students are expected to keep their own grade records as the cumulative average reported in Canvas may be inaccurate at times.

### Gender Inclusive Language Policy:

Please be aware of Wheaton College's policy on inclusive language, "For academic discourse, spoken and written, the faculty expects students to use gender inclusive language for human beings."

### Late Assignment Policy:

Assignments must be turned in at the beginning of class on the due date. Assignments, project deliverables, assigned readings and reflection questions must be completed and submitted by the due date and time. Late assignments will not be accepted unless an extension has been pre-approved by the instructor.

### Academic Integrity Policy:

The Wheaton College Community Covenant, which all members of our academic community affirm, states that, "According to the Scriptures, followers of Jesus Christ will...be people of integrity whose word can be fully trusted (Psalm 15:4; Matt. 5:33-37)." It is expected that Wheaton College students, faculty and staff understand and subscribe to the ideal of academic integrity and take full personal responsibility and accountability for their work. Wheaton College considers violations of academic integrity a serious offense against the basic meaning of an academic community and against the standards of excellence, integrity, and behavior expected of members of our academic community. Violations of academic integrity break the trust that exists among members of the learning community at Wheaton and degrade the College's educational and research mission. See the [Wheaton College Policy on Academic Integrity](#) for details.

### Title IX and Mandatory Reporting:

Wheaton College instructors help create a safe learning environment on our campus. Each instructor in the college has a mandatory reporting responsibility related to their role as a faculty member. Faculty members are

required to share information with the College when they learn of conduct that violates our Nondiscrimination Policy or information about a crime that may have occurred on Wheaton College's campus. Confidential resources available to students include Confidential Advisors, the Counseling Center, Student Health Services, and the Chaplain's Office. More information on these resources and College Policies is available <http://www.wheaton.edu/equityandtitleIX>.

### Use of Artificial Intelligence:

AI Writing tools are not permitted for any stage or phase of work in this class. If you use these tools, your actions would be considered academically dishonest and a violation of the Wheaton College Policy on Academic Honesty.

### Learning and Accessibility Services:

Wheaton College is committed to providing access and inclusion for all persons with disabilities, inside and outside the classroom. Students are encouraged to discuss with their professors if they foresee any disability-related barriers in a course. Students who need accommodations in order to fully access this course's content or any part of the learning experience should connect with Learning and Accessibility Services (LAS) as soon as possible to request accommodations <http://wheaton.edu/las> (Student Services Building - Suite 209, [REDACTED] phone [REDACTED])

hesitate to reach out or ask any questions.

### Course Schedule:

The following schedule is tentative and is subject to change.

| Week | Date   | Reading    | Topic                                                                                                | Deliverables          |
|------|--------|------------|------------------------------------------------------------------------------------------------------|-----------------------|
| 1    | Jan 13 | 12.1       | Introductions, Expectations, and Objectives                                                          |                       |
|      | Jan 15 | 12.1-12.2  | Position, Velocity, Acceleration Vectors, Straight-Line Motion (Constant Acceleration) GUEST LECTURE |                       |
|      | Jan 17 | 12.2-12.3  | Straight-Line Motion (Changing Acceleration)                                                         |                       |
| 2    | Jan 20 |            | No class, MLK day.                                                                                   |                       |
|      | Jan 22 | 12.3       | Erratic Motion                                                                                       |                       |
|      | Jan 24 | 12.3-12.4  | Review erratic motion, Working Groups                                                                | HW #1 Due<br>JC Topic |
| 3    | Jan 27 | 12.4-12.5  | 12.4-12.5 Curvilinear Motion - Cartesian Components                                                  | JC Topic Due          |
|      | Jan 29 | 12.6       | Projectile Motion – GUEST LECTURE                                                                    |                       |
|      | Jan 31 | 12.6       | JC Presentations, Review of Coordinate Motion                                                        | JC Presentations      |
| 4    | Feb 3  | 12.7       | Normal, Tangential Components                                                                        | HW#2 due              |
|      | Feb 5  | 12.8       | Polar and Cylindrical Coordinates                                                                    |                       |
|      | Feb 7  |            | Working Groups                                                                                       |                       |
| 5    | Feb 10 | 12.9-12.10 | Relative Motion                                                                                      | HW #3 Due             |
|      | Feb 12 | Review     | <b>Exam 1 Review Day: Problem-solving</b>                                                            |                       |
|      | Feb 14 |            | <b>Unit Exam 1</b>                                                                                   | HW#4 Due              |
| 6    | Feb 17 |            | No class, President's Day                                                                            |                       |
|      | Feb 19 | 13.1-13.3  | F=ma Straight Line Motion                                                                            |                       |
|      | Feb 21 | 13.4-13.5  | F=ma Normal and Tangential Components                                                                |                       |

|             |           |           |                                                              |                                 |
|-------------|-----------|-----------|--------------------------------------------------------------|---------------------------------|
| 7           | Feb 24    | 13.6-13.7 | F=ma Polar and Cylindrical Coordinates                       | Concept Sketch Due              |
|             | Feb 26    | 14.1-14.2 | Work and Kinetic Energy                                      | HW#5 Due                        |
|             | Feb 28    | 14.2-14.4 | Energy and Power                                             |                                 |
| 8           | Mar 3     | 14.4-14.6 | Energy and Power, Energy Conservation                        |                                 |
|             | Mar 5     |           | Meet with Project Teams                                      | HW#6 Due                        |
|             | Mar 7     | 15.1-15.2 | Review Ch.14, Impulse and Momentum                           |                                 |
| 9           | Mar 10-14 |           | No class, Spring Break.                                      |                                 |
| 10          | Mar 17    | 15.3-15.4 | Momentum (Conservation)                                      |                                 |
|             | Mar 19    |           | Conceptual Mockups Due, Exam Review                          | Mockups due                     |
|             | Mar 21    |           | Unit Exam 2                                                  |                                 |
| 11          | Mar 24    | 15.5      | Momentum (Collisions)                                        |                                 |
|             | Mar 26    | 15.5-15.6 | Impact and Momentum, Angular Momentum                        |                                 |
|             | Mar 28    | 15.6-15.7 | Angular Impulse and Momentum                                 | Project Updates                 |
| 12          | Mar 31    | 16.1-16.2 | Rigid Body Rotation                                          |                                 |
|             | Apr 2     |           | Project Workshop: Prototypes                                 |                                 |
|             | Apr 4     | 16.3-16.4 | Rotation about a Fixed Axis                                  |                                 |
| 13          | Apr 7     | 16.5      | Relative Velocities and Angular Velocity Vector              |                                 |
|             | Apr 9     | 16.5-16.6 | Instantaneous Centers                                        |                                 |
|             | Apr 11    | 16.6-16.7 | Relative Accelerations with IC, Generalized Motion           |                                 |
| 14          | Apr 14    |           | Exam Review and Problem Solving                              |                                 |
|             | Apr 16    |           | Unit Exam 3                                                  | Unit Exam #3                    |
|             | Apr 18    |           | No class, Good Friday.                                       |                                 |
| 15          | Apr 21    |           | Project Workday                                              |                                 |
|             | Apr 23    | 17.1-17.5 | Planar Equations of Motion: Moment of Inertia                |                                 |
|             | Apr 25    | 17.1-17.5 | Planar Equations of Motion: Kinetics                         |                                 |
| 16          | Apr 28    | 17.1-17.5 | Planar Equations of Motion: General Plane                    |                                 |
|             | Apr 30    |           | Final Review                                                 |                                 |
|             | May 1     | Last day  | Final Project Presentations                                  | Final Project Due               |
| Finals Week | May 5     |           | Final Exam<br>Wednesday, May 7 <sup>th</sup> , 10:30-12:30PM | Final Report Due<br>by midnight |